

How comp work

724
↑
→ ones
→ tens
→ thousand

$$\begin{array}{r} 7 \times 100 = 700 \\ 2 \times 10 = 20 \\ 4 \times 1 = 4 \\ \hline 724 \end{array}$$

$$724 = 7 \times 100 + 2 \times 10 + 4 \times 1 \\ = 7 \times 10^2 + 2 \times 10^1 + 4 \times 10^0$$

$$10^0 = 1$$

$$10^1 = 10$$

$$10^2 = 100$$

Decimal $\rightarrow$ base 10

Binary $\rightarrow$ base 2

Binary to decimal

$$\begin{aligned} 1011_2 &= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 \\ &= 8 + 0 + 2 + 1 \end{aligned}$$

$$1011_2 = 11_{10}$$

$$1_{10} = 2^0$$

$$2_{10} = 2^1$$

$$3_{10} = 2^1 + 2^0$$

$$4_{10} = 2^2$$

$$5_{10} = 2^2 + 2^0$$

$$= 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 101_2$$

$$13_{10} = 2^3 + 2^2 + 2^0$$

$$= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 1101_2$$

$$\begin{array}{r} 10110 \\ 1001 \end{array} \left\} \underline{22+9=31}\right.$$

↓

$$\begin{array}{r} 10110 \\ 01001_2 \\ \hline 11111_2 \\ \hline \end{array}$$

↓

$$\rightarrow 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ = 31_{10}$$

$$\begin{array}{r} 161 \\ 111 \\ \hline 1100 \end{array}$$

$$\left\{ \begin{array}{ccccc} 0 & 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 1 \\ \hline & 1 & 0 & 10 & 11 \end{array} \right\}$$

1) D → B

2) B → D

3) Add Binary

4) Sub binary

"two complement"

- Every number binary shall use 4 digits

$$\begin{aligned} 1111_2 &= -1(2^3) + 2^2 + 2^1 + 2^0 \\ &= -8 + 4 + 2 + 1 \\ &= -1 \end{aligned}$$

$$\begin{aligned} 1011_2 &= -1(1 \times 2^3) + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= -8 + 0 + 2 + 1 \\ &= -8 + 2 + 1 \\ &= -5 \end{aligned}$$

$$-5 + -1 = -6 \rightarrow$$

$$= \boxed{11010} = -6$$

$$\begin{array}{r} 111 \\ 1011 \\ \hline 1111 \\ \hline 11010 \end{array}$$

To negate a number
 1) invert all the digit position
 2) add 1 to result

$$-6_{10} = 1010_2$$

$$1) 0101$$

$$\begin{array}{r} 2) 0101 \\ 0001 \\ \hline 0110 \end{array}$$

$$= 2^2 + 2^1$$

$$= 4 + 2 = 6$$

$$9 - 6 = ?$$

$$6_{10} = 0110_2$$

$$2_{10} = 0010_2$$

~~2~~

$$\begin{array}{r} -2_{10} = 1101 \\ 0001 \\ \hline 1110 \end{array}$$

$$6 + (-2)$$

$$\begin{array}{r} 11 \\ 0110 \\ 1110 \\ \hline 10100 \\ 44 \end{array}$$

$$6 - 4 = 2$$